

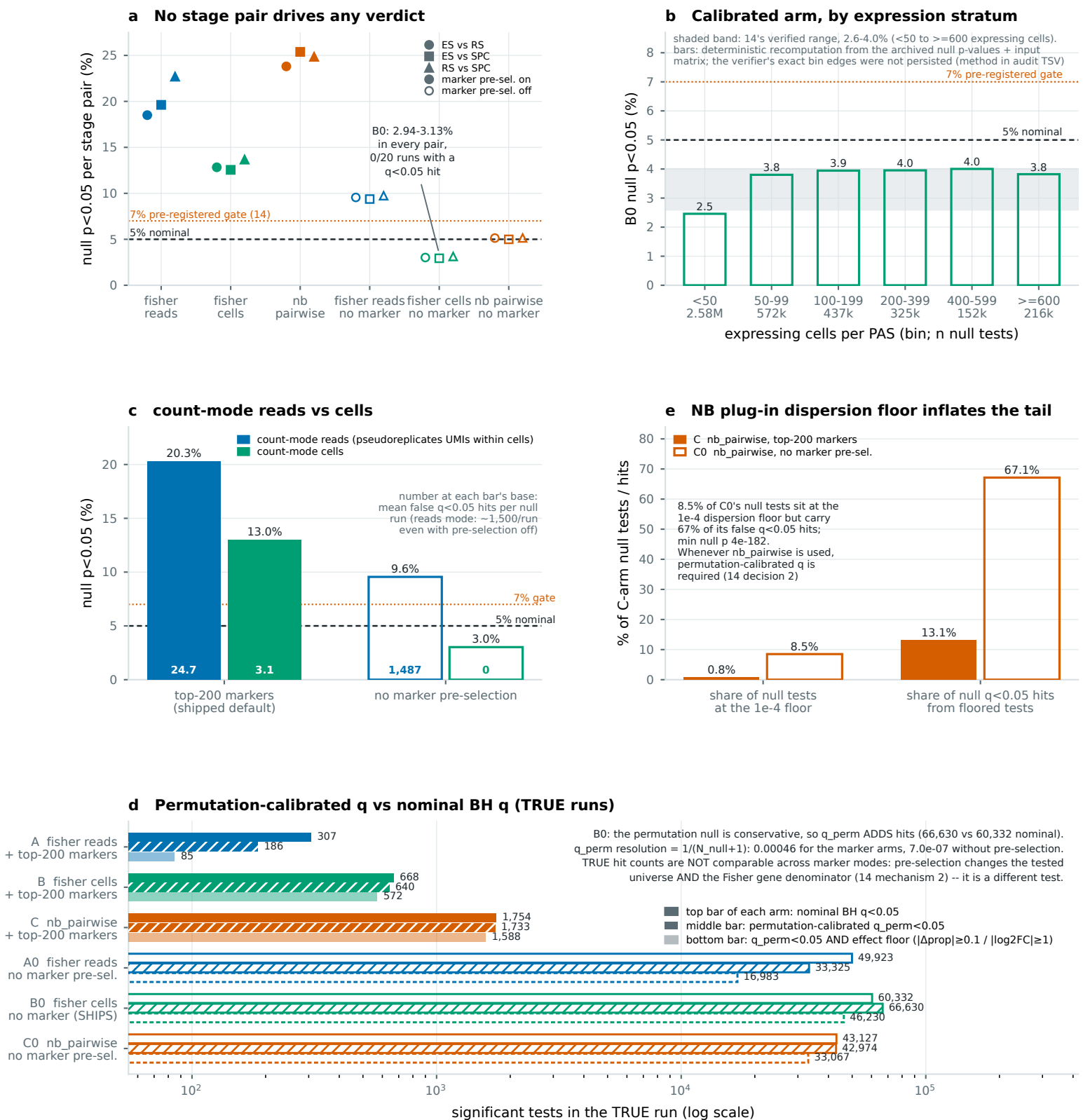


Fig S9 | Calibration deep-dive behind Fig 4 (fdr_calibration_v2 run; testis mouse 1, 1,230 cells, 76,711 PAS, 3 stage pairs, 20 shared label permutations; manuscript/14, verified SOUND). (a) Null $p < 0.05$ per stage pair for all six switch-test configurations: no pair drives any verdict; the shipping configuration B0 (Fisher, count-mode cells, no marker pre-selection) is 2.94-3.13% with 0/20 null runs carrying any $q < 0.05$ hit. (b) B0 stratified by a label-independent expression measure: every stratum is below the 5% nominal rate -- the test is never anti-conservative at any expression level. 14's verified range (2.6-4.0%) is the shaded band; the bars recompute it from the archived per-test null p-values and input matrix under the declared binning (2.5-4.0%; the verifier's exact bin edges were not persisted (method in audit TSV)). (c) count-mode reads pseudoreplicates UMIs within cells: 20.3% vs 13.0% null $p < 0.05$ with marker pre-selection on, 9.6% vs 3.0% with it off, and ~1,500 false $q < 0.05$ hits per null run in reads mode even without pre-selection. (d) Permutation-calibrated q (empirical $p = \text{rank in the pooled same-pair null, BH per pair}$) vs nominal BH q on the TRUE runs; for B0 the permutation null is conservative and q_{perm} adds hits. (e) nb_pairwise's plug-in dispersion floor (1e-4): 8.5% of C0's null tests carry 67% of its false $q < 0.05$ hits (13% in C; min null p 4e-182) -- whenever marker pre-selection or nb_pairwise is used, permutation-calibrated q is required (14 decision). Permutation-harness cost (fig4_calibration_stats.tsv): TRUE runs 9-310 s per arm, mean permutation run 8-774 s (20 permutations per arm, shared seeds). Caveats travel from 14: single mouse, single tissue, very large true stage effects; the label-permutation null tests exchangeability only. Sources: fig4_calibration_stats.tsv, fig4_calibration_per_pair.tsv, results/fdr_calibration_v2/ (report.json; null p-values gz; input h5ad); no permutation was re-run. Every plotted value: results/figures/manuscript/figS9_calibration_extended_*.tsv.